

Principles of Oral Surgery

OMFS 711

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Objectives & Competency Statements

Learning objectives

- **Perform a thorough pre-operative assessment including appropriate consultations, medical risk assessment, premedication, and incorporating various modifications to treatment. (competencies 2,3,4,5,6,8)**
- **Identify indications and contraindications and procedures for surgical extraction of erupted, partially, and fully impacted third molars. (competencies 2,3,4,5,6,8)**
- **Provide sound post-surgical management and effectively evaluate and treat post-operative complications (competencies 2,3,4,5,6,8)**

Pain and Anxiety Control in Surgical Practice

It is incumbent upon those performing oral surgery to master techniques that will control perioperative pain and anxiety. For most routine oral surgical procedures, local anesthesia is sufficient to manage pain produced during surgery and for the early postoperative period.

Anxiety control, on the other hand, is a more complex topic. Patients may or may not require pharmacologic control. Most patients are adequately managed using nonpharmacologic behavioral techniques. However, some form of drug-mediated sedation is often useful or required for patients with higher levels of presurgical anxiety or for procedures known to cause intraoperative anxiety

Local Anesthesia

The ability to locally anesthetize a specific part of the body is one of the marvels of pharmacology. It is hard to imagine the modern practice of dentistry without the availability of local anesthetics (LAs). The dental pulp and soft tissues adjacent to teeth are highly sensitive to stimulation of all forms, including pain-provoking stimuli. Thus profound local anesthesia is mandatory to properly perform oral surgery on an awake patient.

A large number of LAs are available for use by dental practitioners. However, like most medications, those administering drugs find it optimal to limit the number of different forms of a drug to produce a similar effect. This gives them the opportunity to gain valuable clinical experience with the use of these drugs, enabling them to recognize usual and unusual patient reactions and more easily stay abreast of new knowledge related to these drugs.

Mechanism of Action

LAs are, by definition, designed to block the function of sensory nerves, although they are also able to inhibit motor nerves and other nerve tissues. LAs primarily function by raising the membrane firing threshold necessary to trigger or propagate an electric impulse. LAs also produce local anesthesia by affecting sensory receptors and nerve membranes in other ways. The end result is that the nerve membrane remains in a polarized state unable to conduct impulses and not transmit otherwise painful sensations.

Duration of Anesthesia

Local Anesthetic	Maxillary Teeth	Mandibular Teeth	Soft Tissue
Group 1 ^a	10–20?min	40–60?min	2–3?h
Group 2 ^b	50–60?min	90–100?min	3–4?h
Group 3 ^c	60–90?min	3?h	4–9?h

Recommended Maximum Local Anesthetic Doses

Drug/Solution	Maximum Amount (mg/kg)	Number of Cartridges for 70-kg (154-lb) Adult	Number of Cartridges for 20-kg (44-lb) Child
Lidocaine 2% with 1 : 100,000 epinephrine	5.0	10	3.0
Mepivacaine 2% with 1 : 20,000 levonordefrin	5.0	10	3.0
Mepivacaine 3% (no vasoconstrictor)	5.0	6	2.0
Prilocaine 4% with 1 : 200,000 epinephrine	5.0	6	2.0
Articaine 4% with 1 : 100,000 epinephrine	7.0	6	1.5
Bupivacaine 0.5% with 1 : 200,000 epinephrine	1.5	10	3.0
Etidocaine 1.5% with 1 : 200,000 epinephrine	8.0	15	5.0

Vasoconstrictors

The two most common vasoconstrictors added to LAs used for dental surgery are epinephrine and levonordefrin. LAs used for oral surgery have varying concentrations of these two drugs. Epinephrine is added to all the LAs discussed in this chapter except mepivacaine. Mepivacaine is available in two forms; 3% mepivacaine used for dentistry contains no vasoconstrictor, whereas the 2% formulation of mepivacaine has levonordefrin in a 1/20,000 concentration.

Both epinephrine and levonordefrin prolong the duration of local anesthesia by producing local vasoconstriction. They also can promote local hemostasis through their vasoconstrictive effects on capillary beds. The cardiac effects increase myocardial oxygen consumption and may provoke dysrhythmias, therefore techniques to limit the amount of these vasoconstrictors are part of the standard protocol of their administration; namely, aspirating before depositing the anesthetic in tissues with sizable blood vessels and limiting the total amount of local anesthetic used.

Modulation of Injection Discomfort

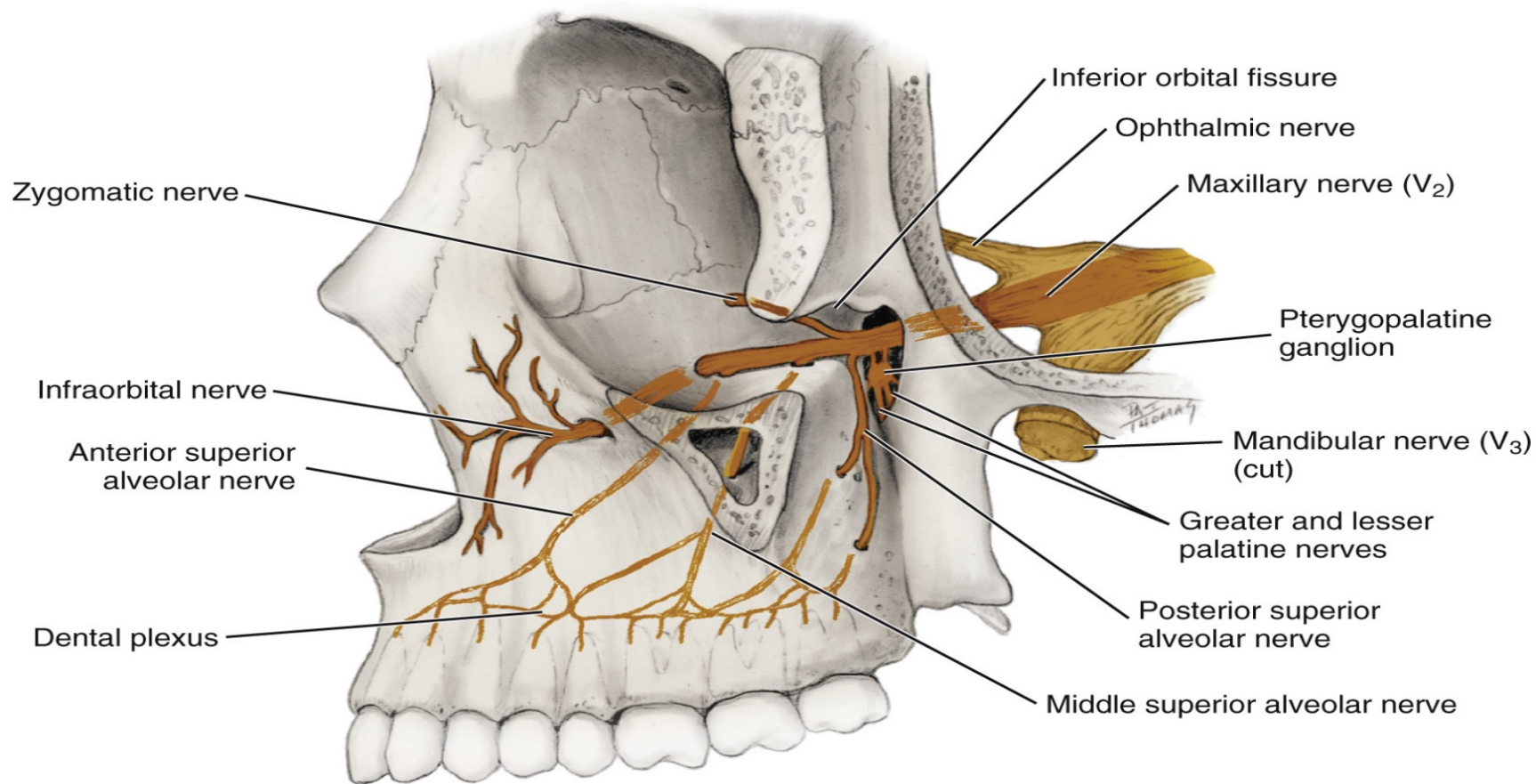
Benzocaine has pharmacologic properties that make it a useful topical anesthetic for oral mucosa. It has a very rapid onset of action. When applied to dried mucosa it can eliminate the discomfort of needle insertion within 60 seconds. However, benzocaine does not penetrate deep enough to eliminate the discomfort of anesthetic deposition.

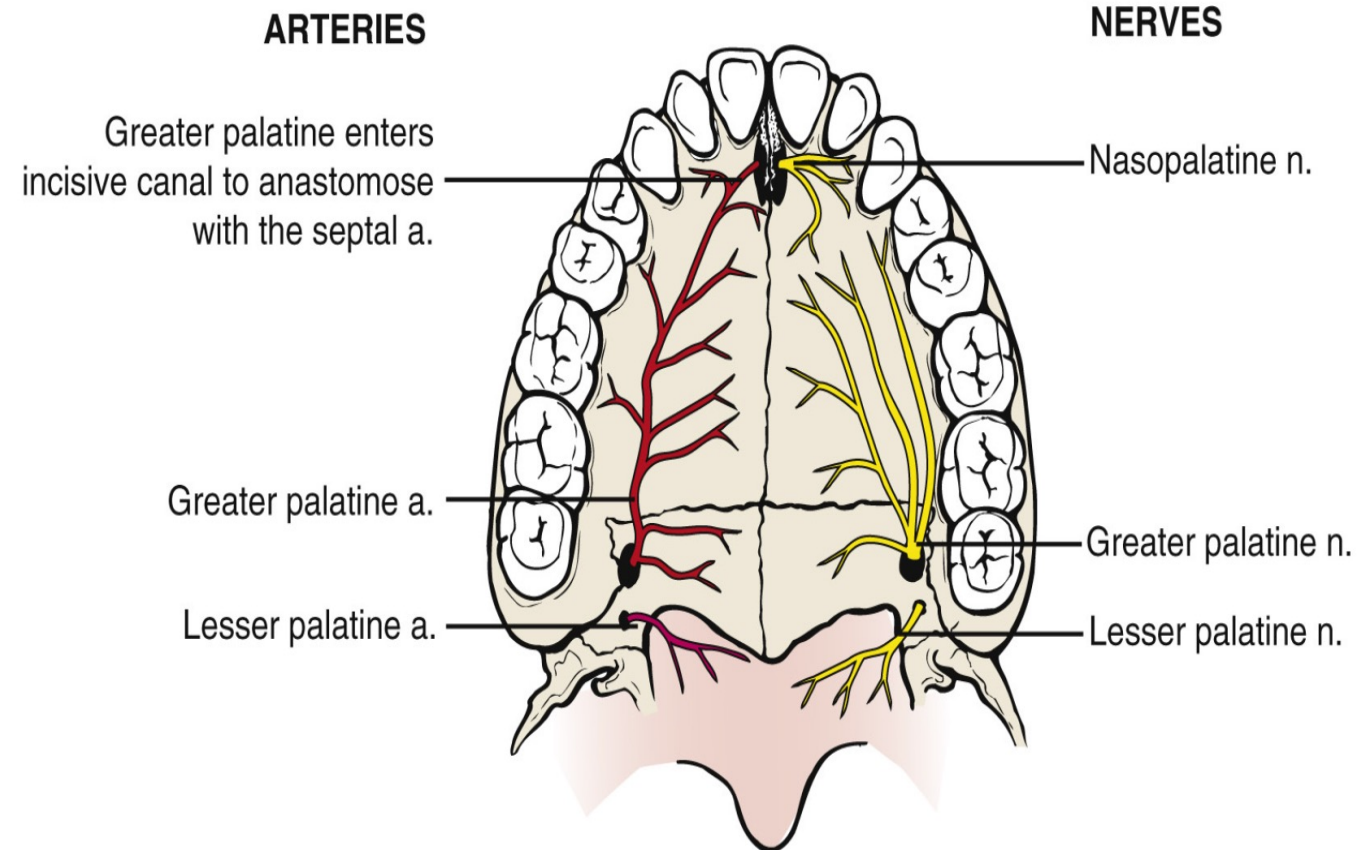
Other approaches to lessening the pain of a local anesthetic injection include lowering the rate of injection, prewarming anesthetic cartridges, and using distraction techniques such as wiggling adjacent tissue like the cheek or talking to the patient about topics unrelated to their surgery during the injection.

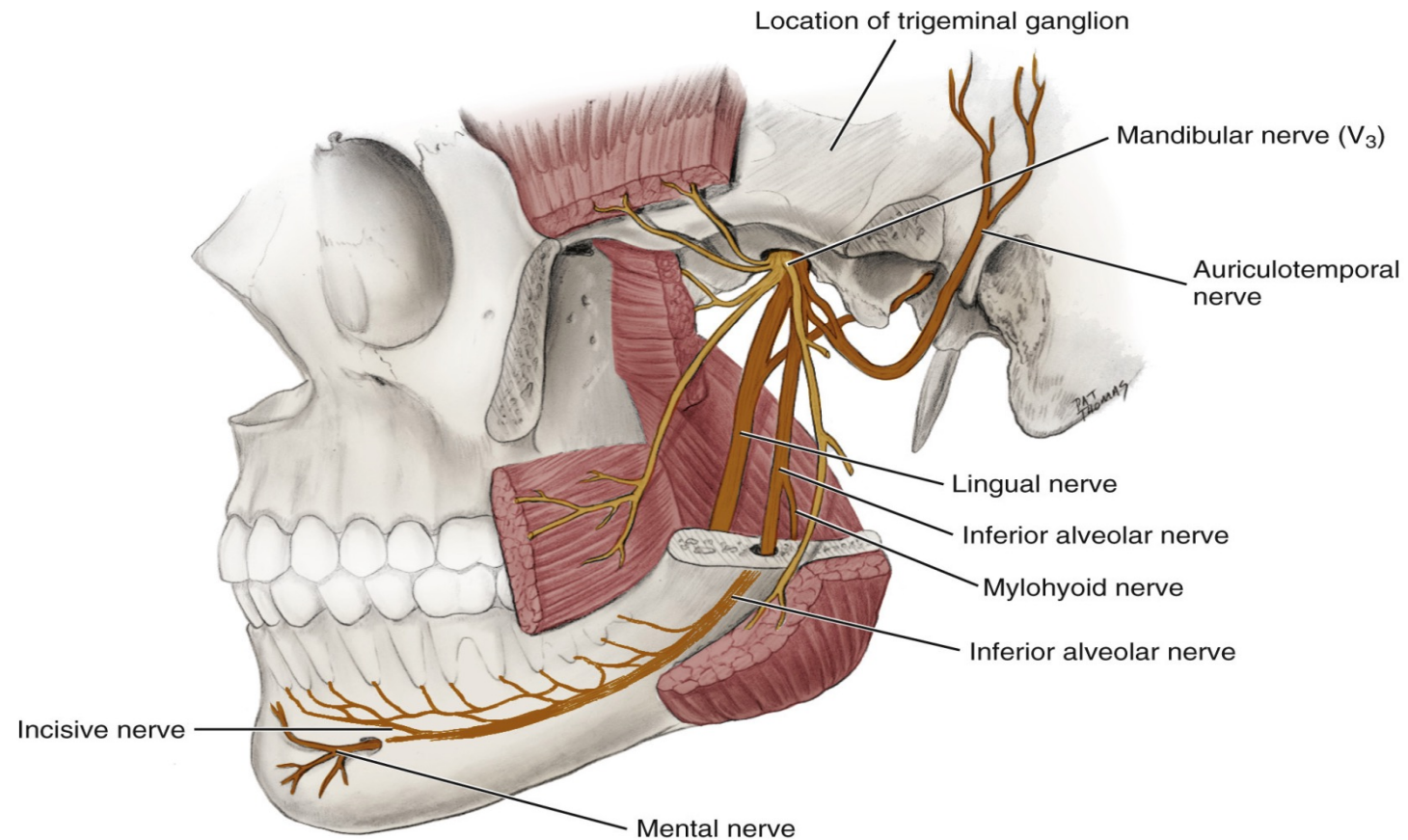
Relevant Anatomy

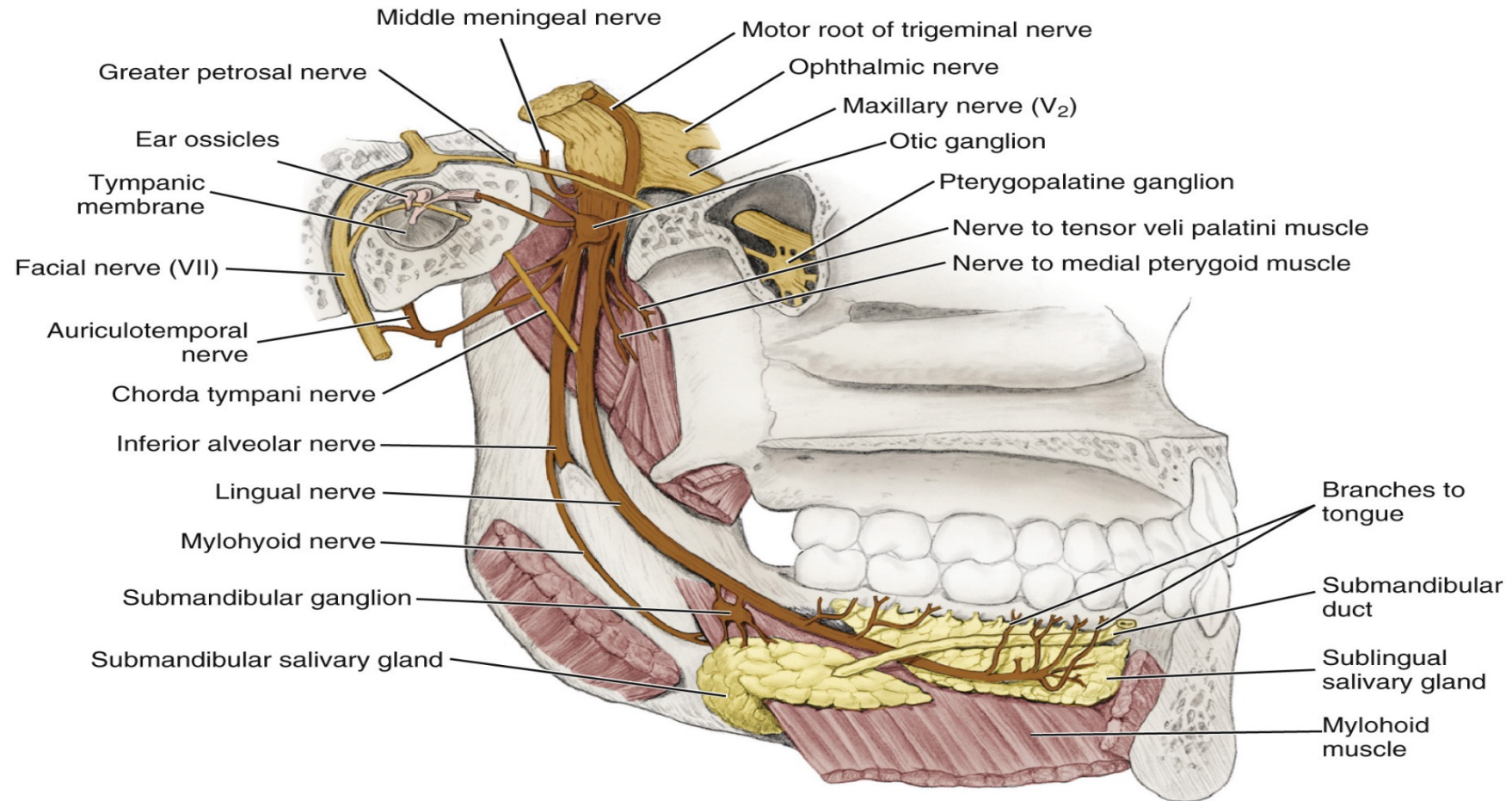
Sensory Innervation of Jaws

Nerve	Teeth	Soft Tissue
Inferior alveolar nerve	All mandibular teeth	Buccal soft tissue of premolars, canines, and incisors
Lingual nerve	None	Lingual soft tissue of all teeth
Long buccal nerve	None	Buccal soft tissue of molars and the second premolar
Anterior superior alveolar nerve	Maxillary incisors and canine teeth	Buccal soft tissue of incisors and canines
Middle superior alveolar nerve	Maxillary premolars and a portion of the first molar tooth	Buccal soft tissue of premolars
Posterior superior alveolar nerve	Maxillary molars except for a portion of the first molar tooth	Buccal soft tissue of molars
Greater palatine nerve	None	Lingual soft tissue of molars and premolars
Nasopalatine nerve	None	Lingual soft tissue of incisors and canines









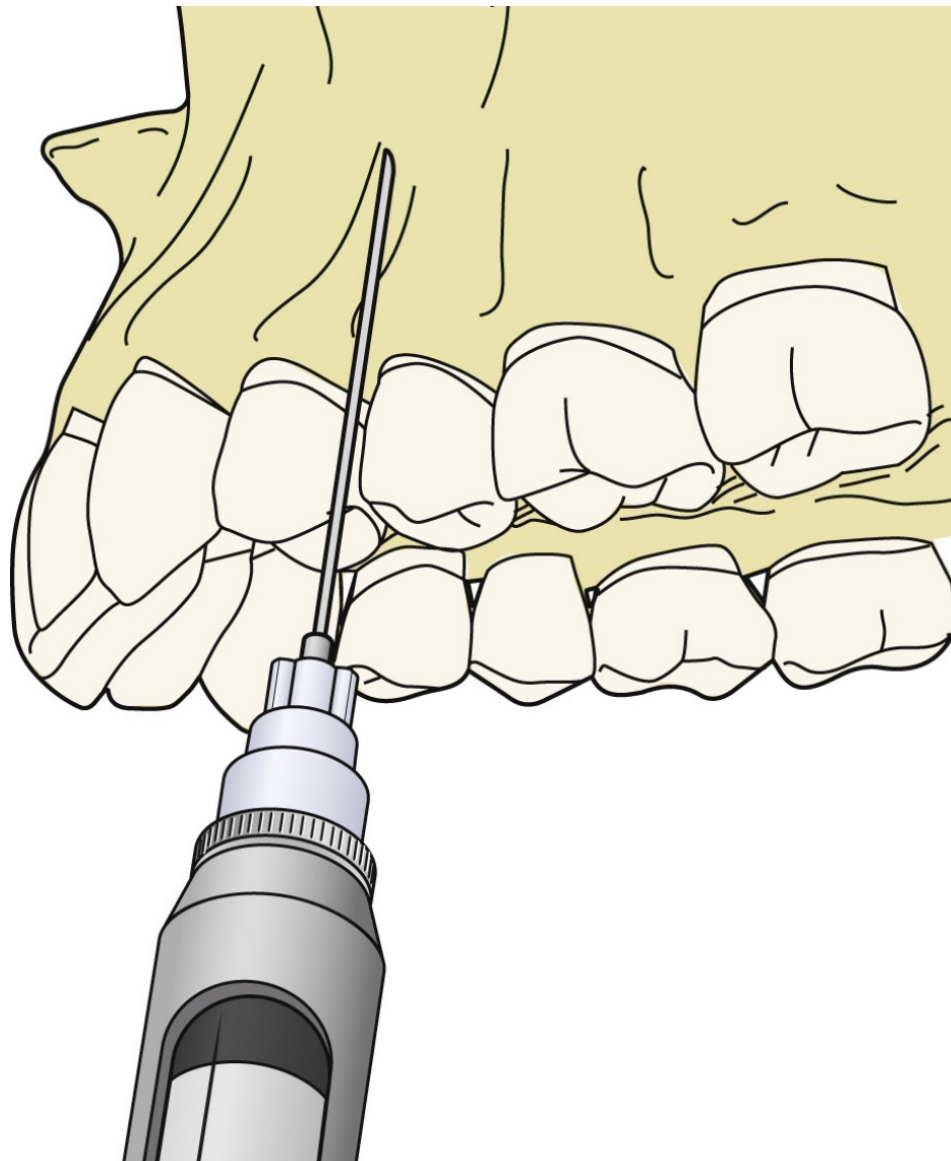
Dense local anesthesia results in the loss of all pain, temperature, and touch sensations, but it does not anesthetize the proprioceptive fibers of the involved nerves. Thus, during an extraction, the patient feels a sensation of pressure, especially when the force is substantial. The surgeon must therefore remember that the patient will need to distinguish between sharp pain and the dull, albeit intense, feeling of pressure when determining the adequacy of anesthesia. It is often difficult to make this distinction.

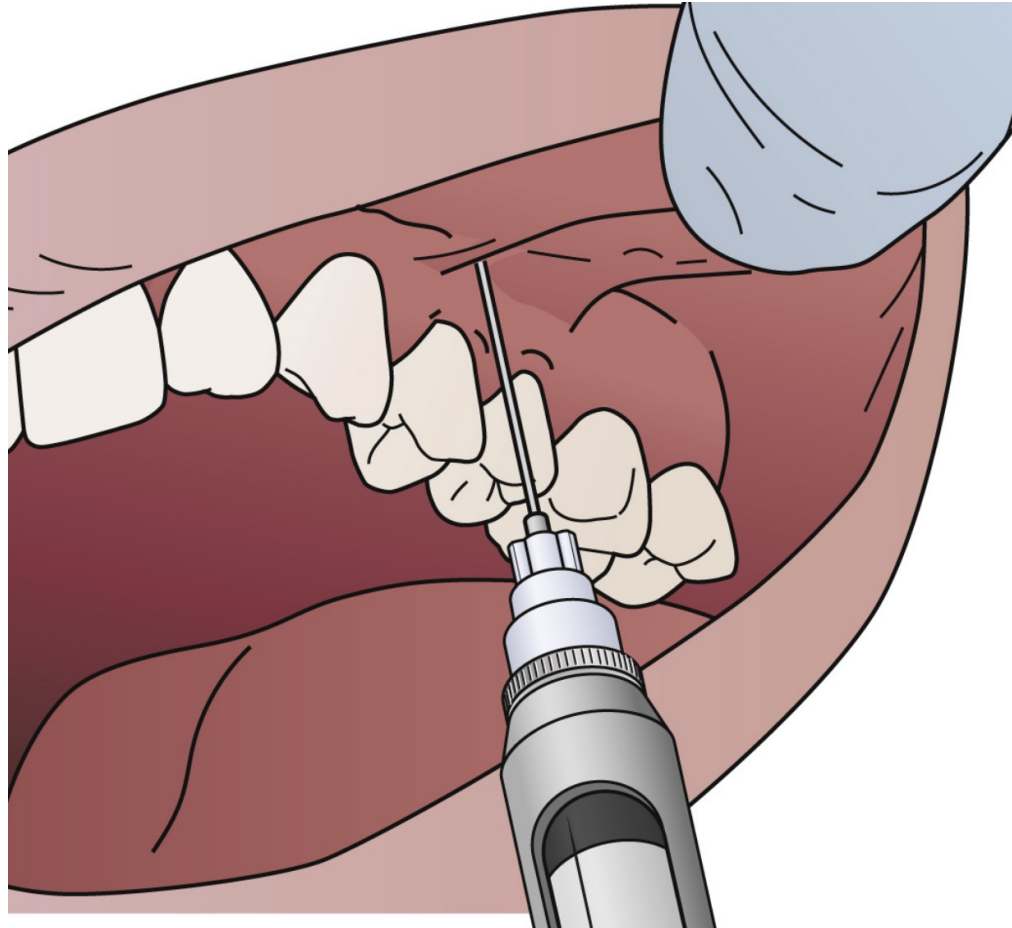
Technique for Administering LA for Oral Surgery

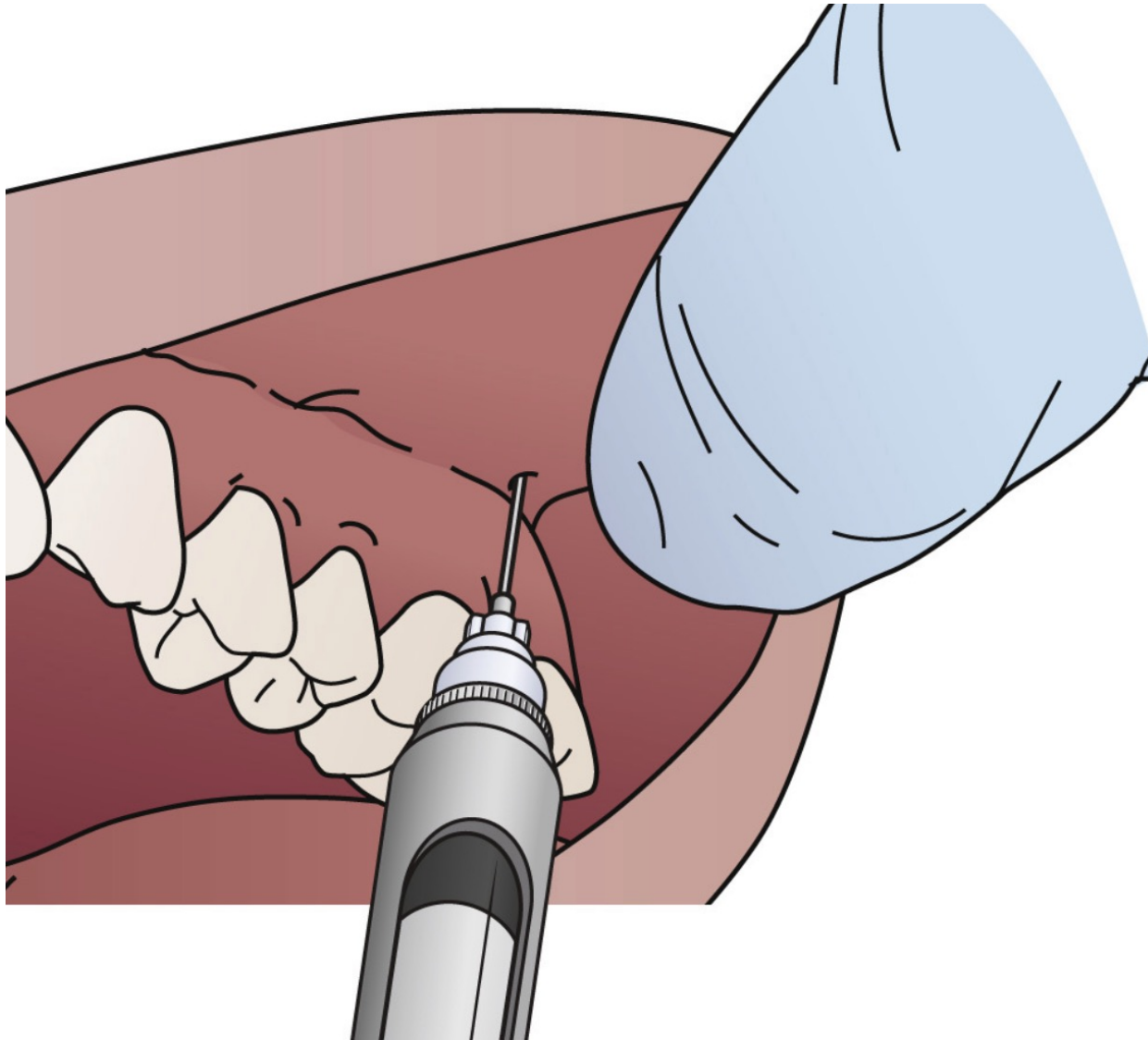
The primary techniques used for dentoalveolar surgery are the inferior alveolar-lingual nerve block and long buccal nerve block in the mandible, whereas in the maxilla the teeth are anesthetized via buccal/labial infiltration along with palatal anesthesia of the incisive and/or greater palatine nerves. In addition, the lingual or palatal soft tissue of a tooth can be effectively anesthetized by infiltration in the soft tissue adjacent to the tooth or teeth to be operated upon. A few caveats apply to all of these types of injection. First, the slower the injection, the less pain produced. However, the slow speed of injection should be balanced against the anxiety most patients suffer as long as the injection is still being performed. Second, injections into palatal tissue tend to be more uncomfortable due to the limited amount of loose connective tissue present.

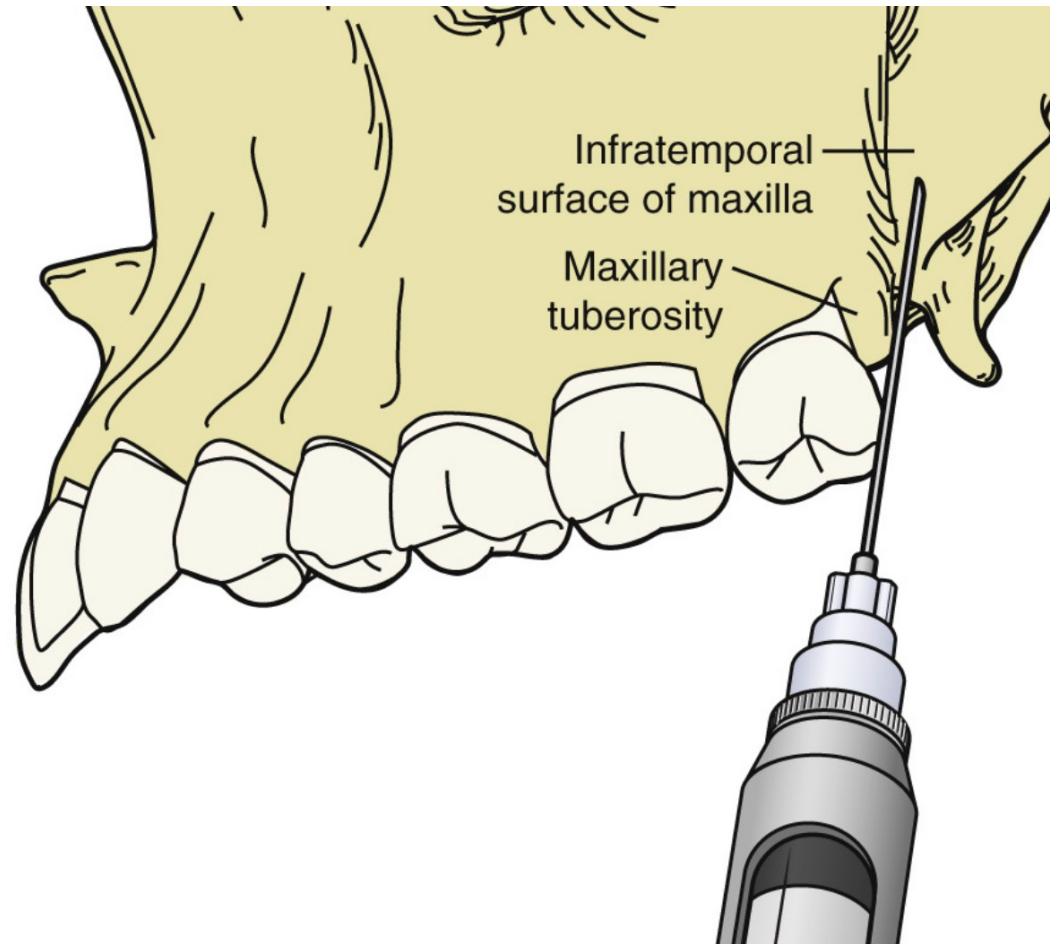
Third, injections in areas with larger blood vessels should be preceded by aspiration to help lessen the possibility of intra-arterial injection. Of the techniques discussed, this applies to the inferior alveolar and posterior superior alveolar nerve blocks. Fourth, the onset of action of injected LA varies based on its pharmacology and the precision of drug deposition, therefore adequate time should be allowed to elapse before commencing surgery.

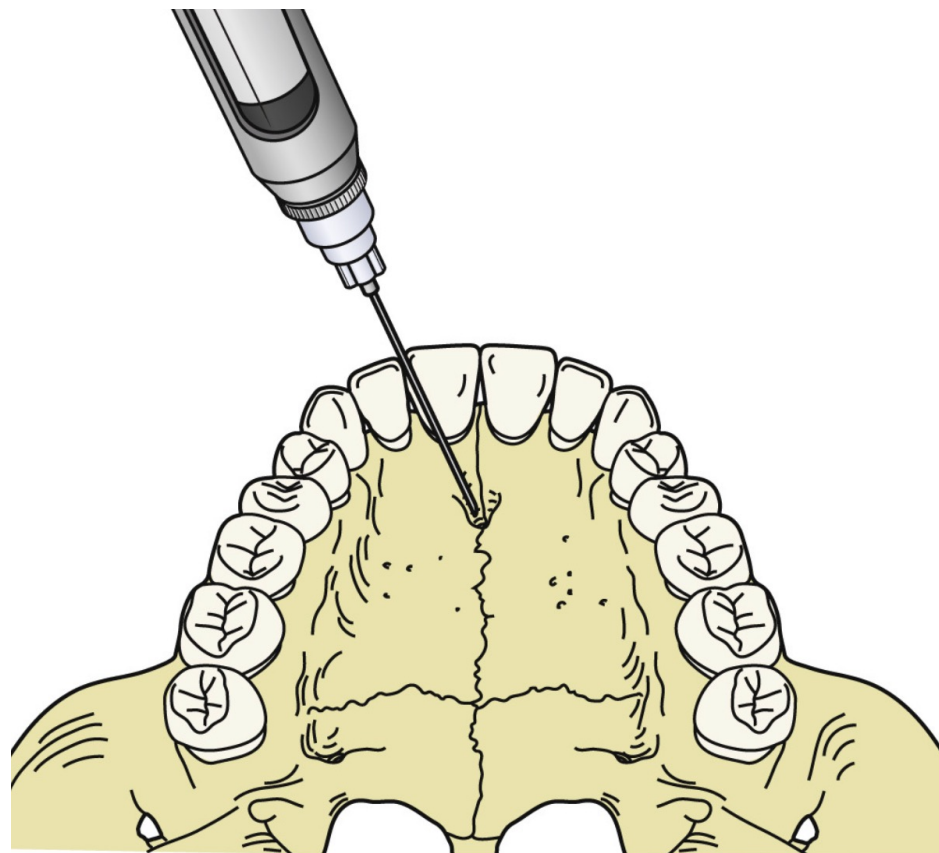
When attempting to anesthetize the inferior alveolar, lingual, and buccal nerves, the area of expected anesthesia should be tested for the patient's ability to detect a painful stimulus before the profound anesthesia is presumed to be present. Simply asking the patients if they feel anesthetized (numb) is not sufficient. Remember that a patient's proprioception is not eliminated by an LA, so both inadequately anesthetized and fully anesthetized patients retain that sensibility.

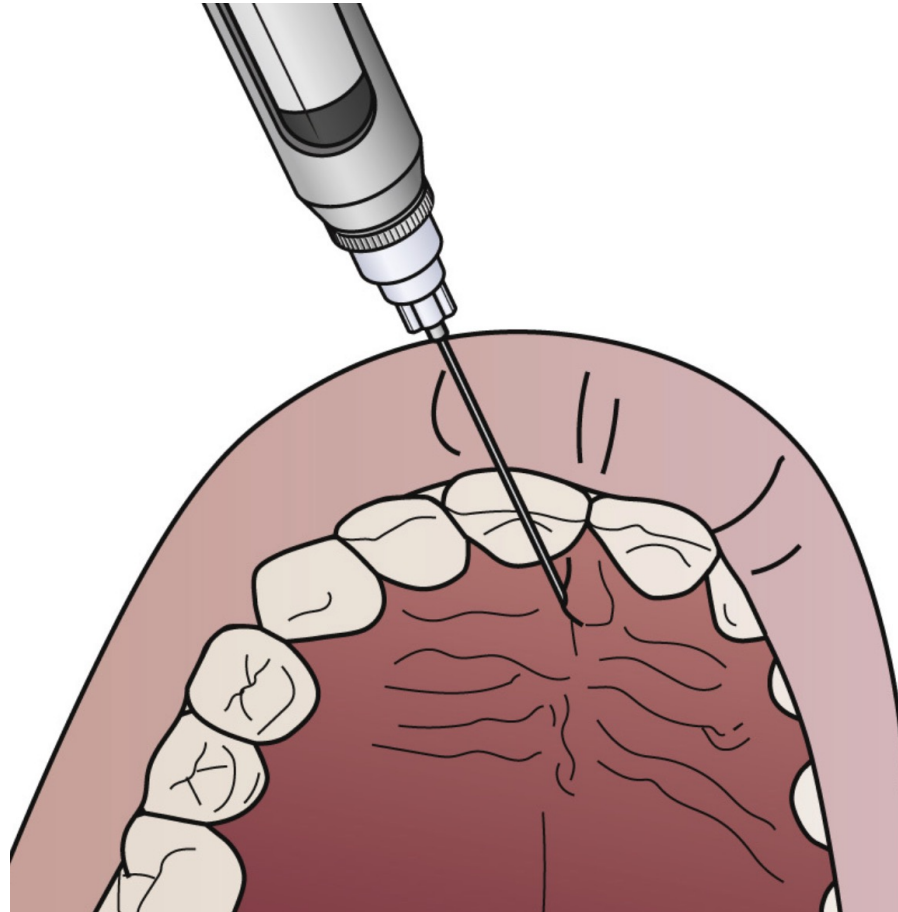


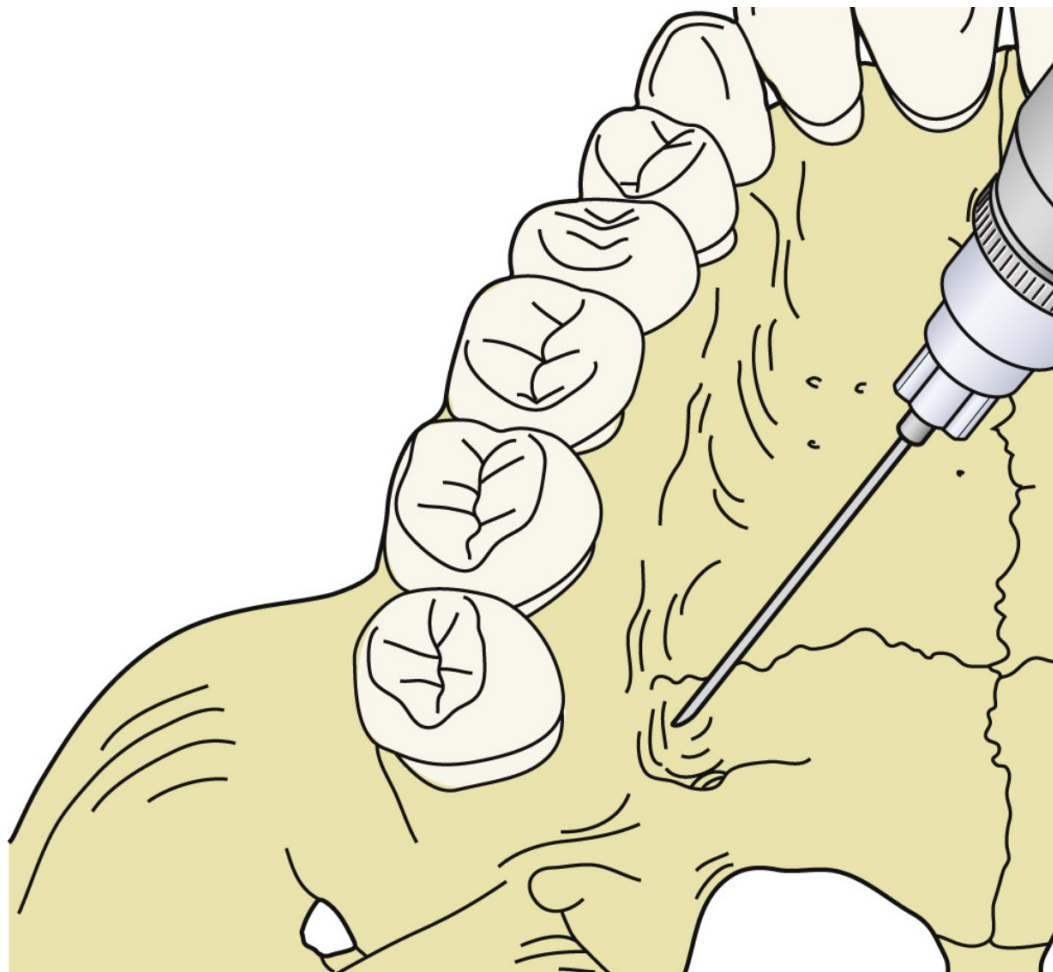


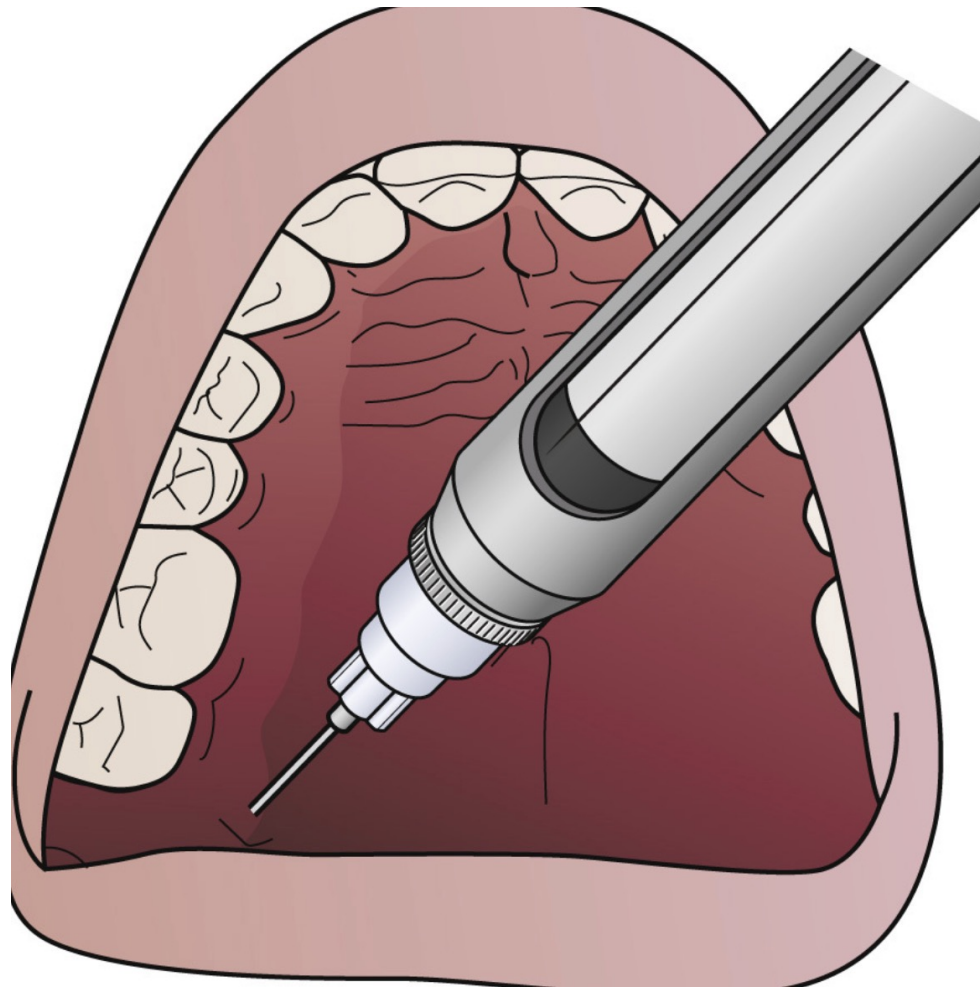






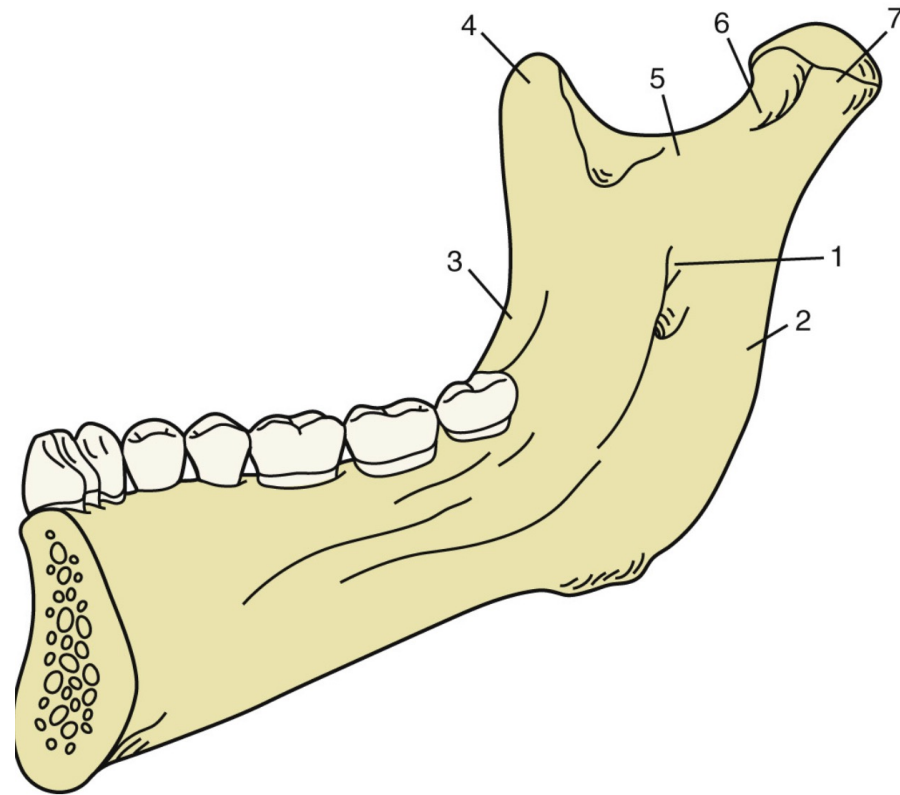


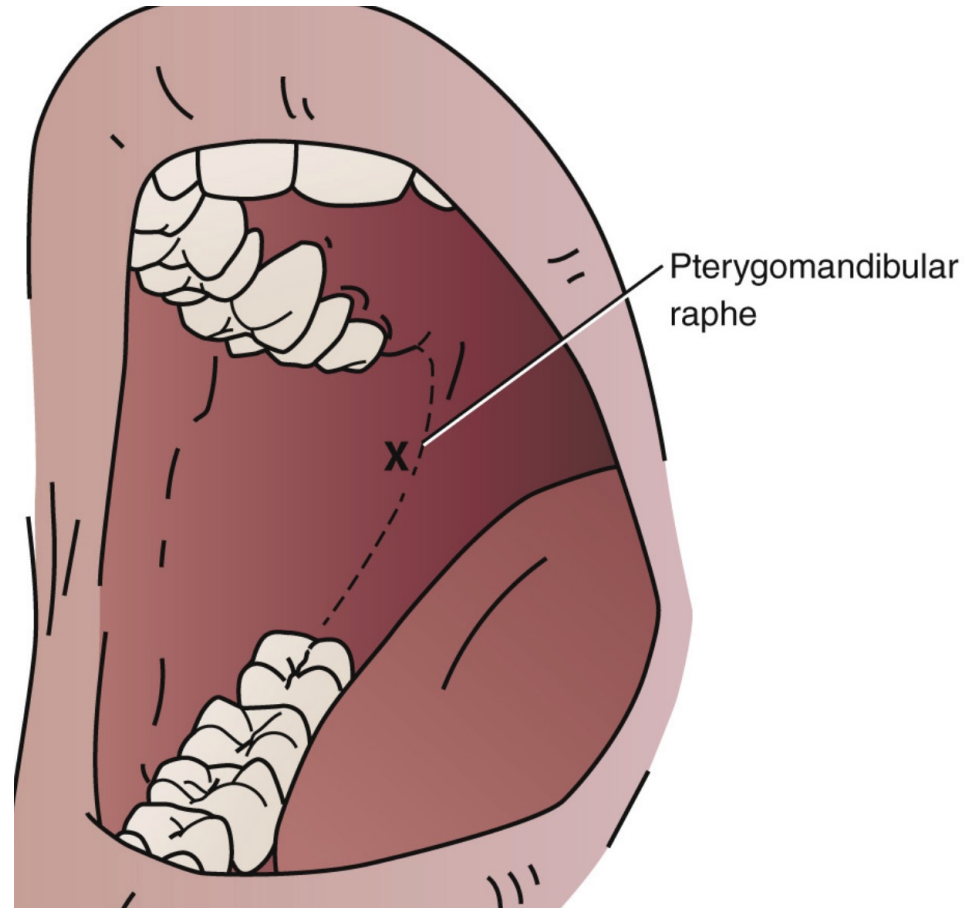


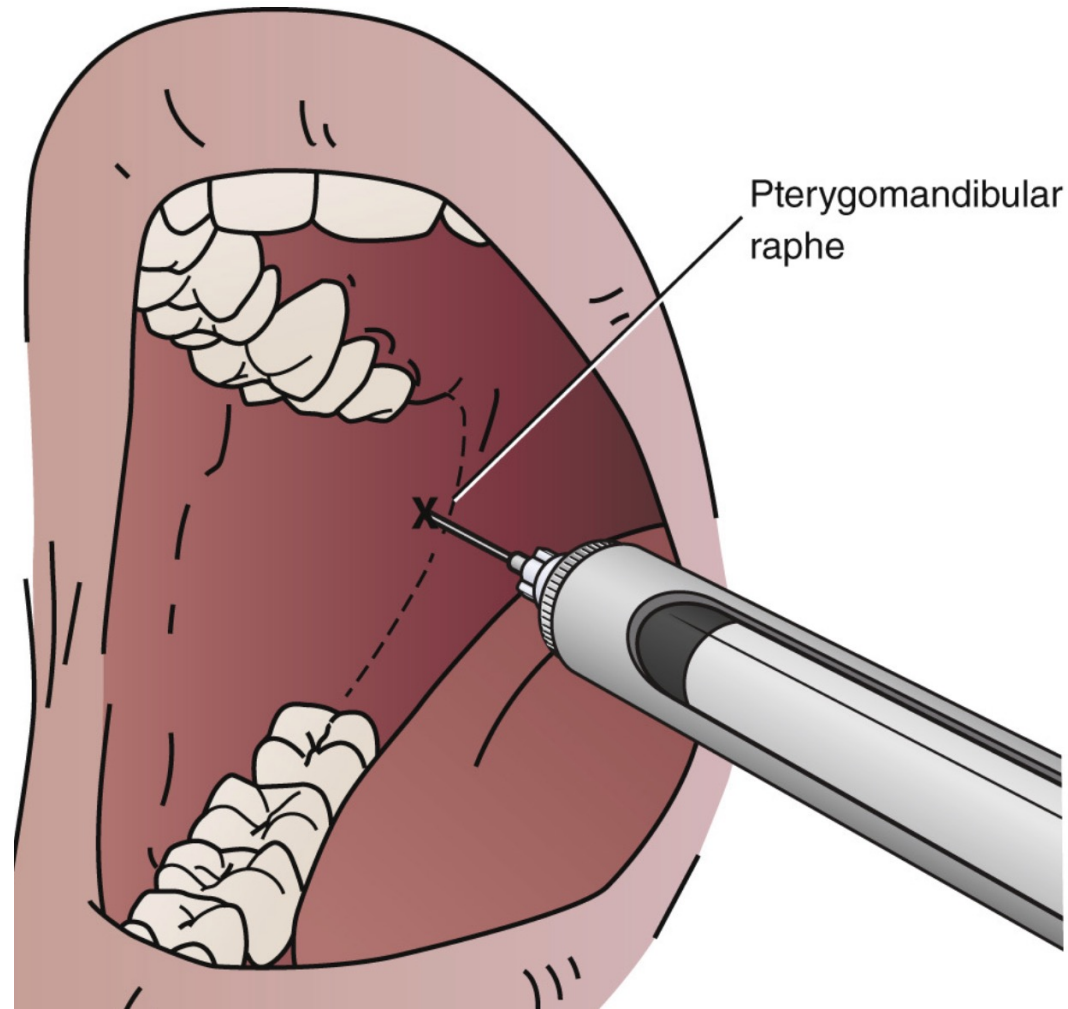


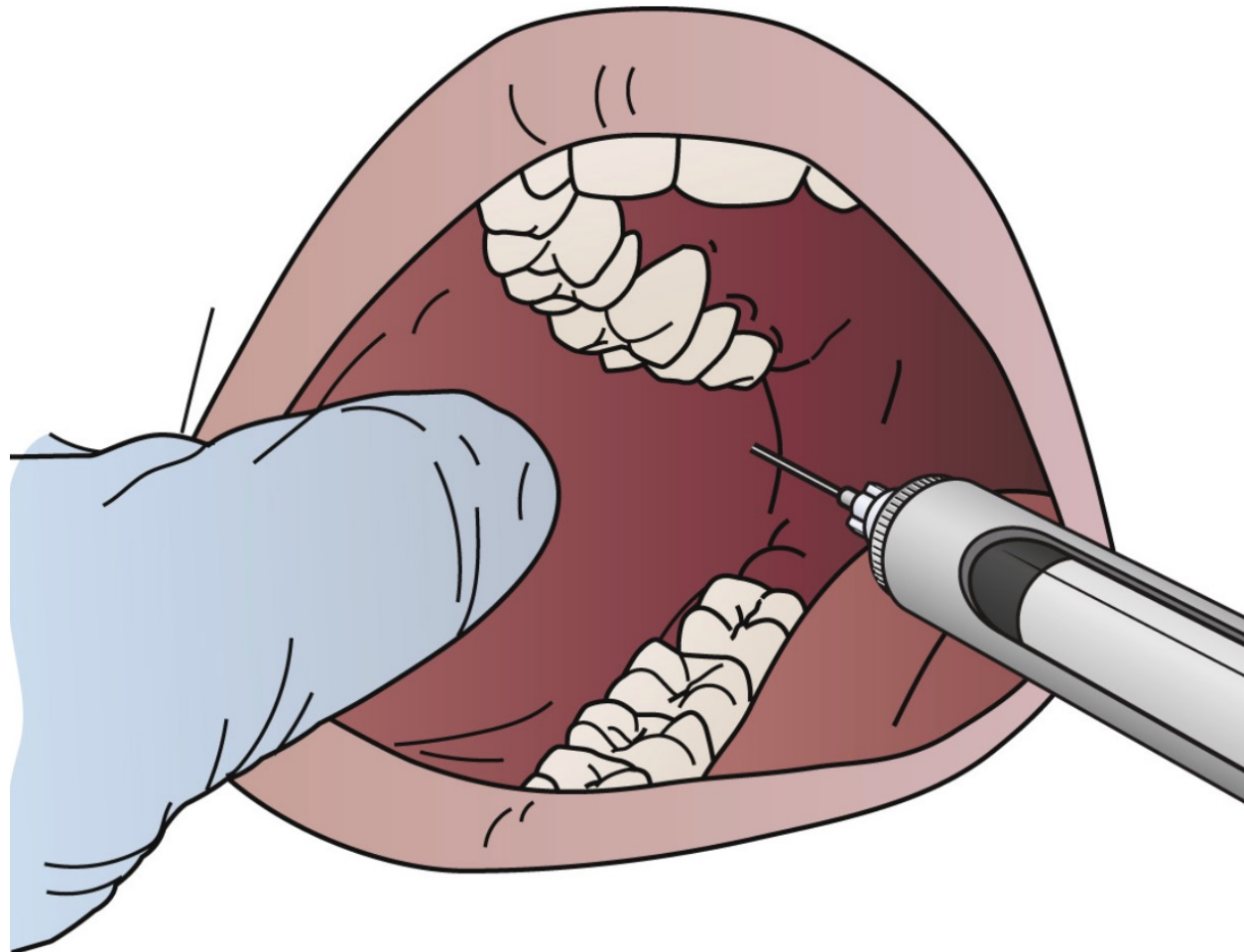
Mandibular Injections

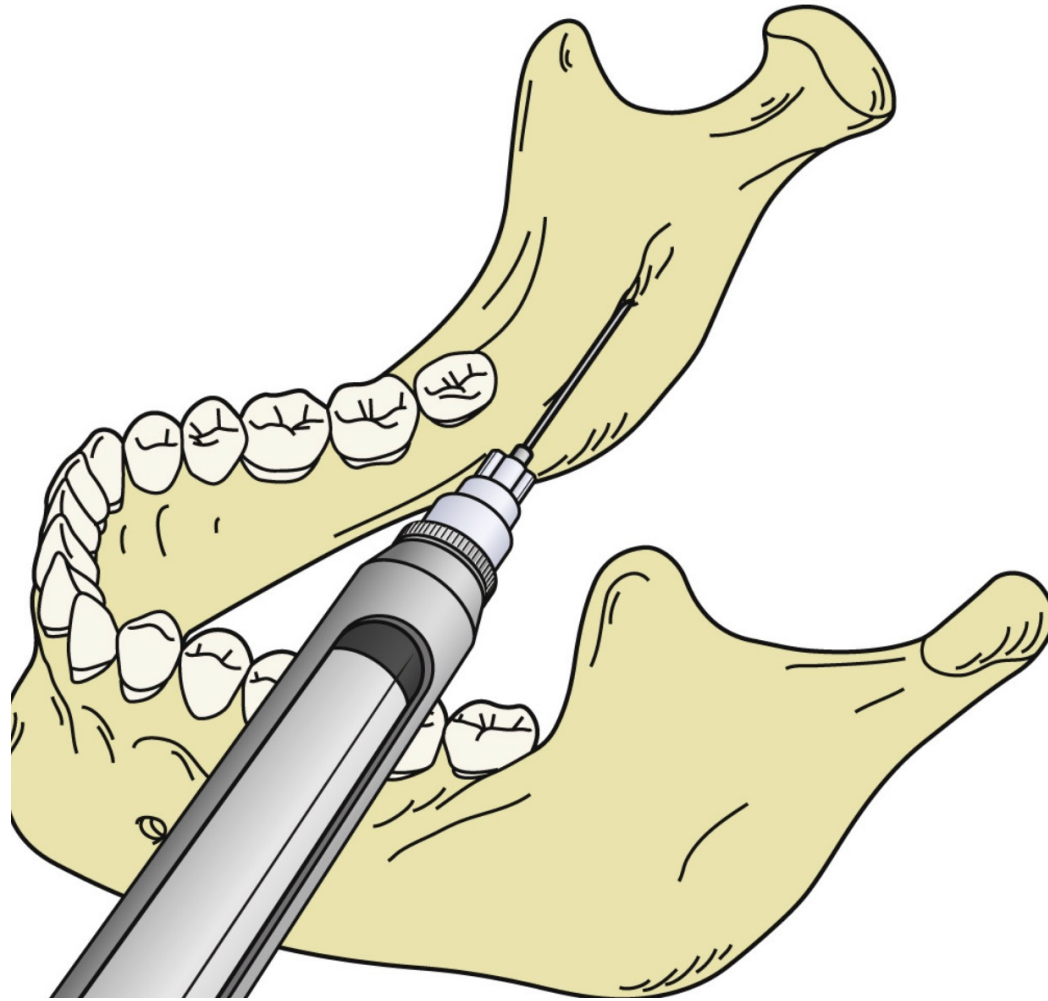
Local anesthesia of mandibular teeth is more complex due to the thickness of the alveolar bone around the apices of those teeth. Individuals desiring profound pulpal anesthesia of anterior mandibular teeth should avoid attempting to do so with a mental nerve block. Mental nerve blocks produce excellent anesthesia of the soft tissues in the distribution of that nerve but rarely provide reliable and adequate pulpal anesthesia. In addition, giving a mental nerve block makes it difficult to then determine the adequacy of a subsequently given inferior alveolar nerve block. Also, as discussed earlier, when performing procedures near the midline there is usually crossover innervation from the inferior alveolar nerve on the contralateral side, so bilateral inferior alveolar nerve blocks may be necessary.



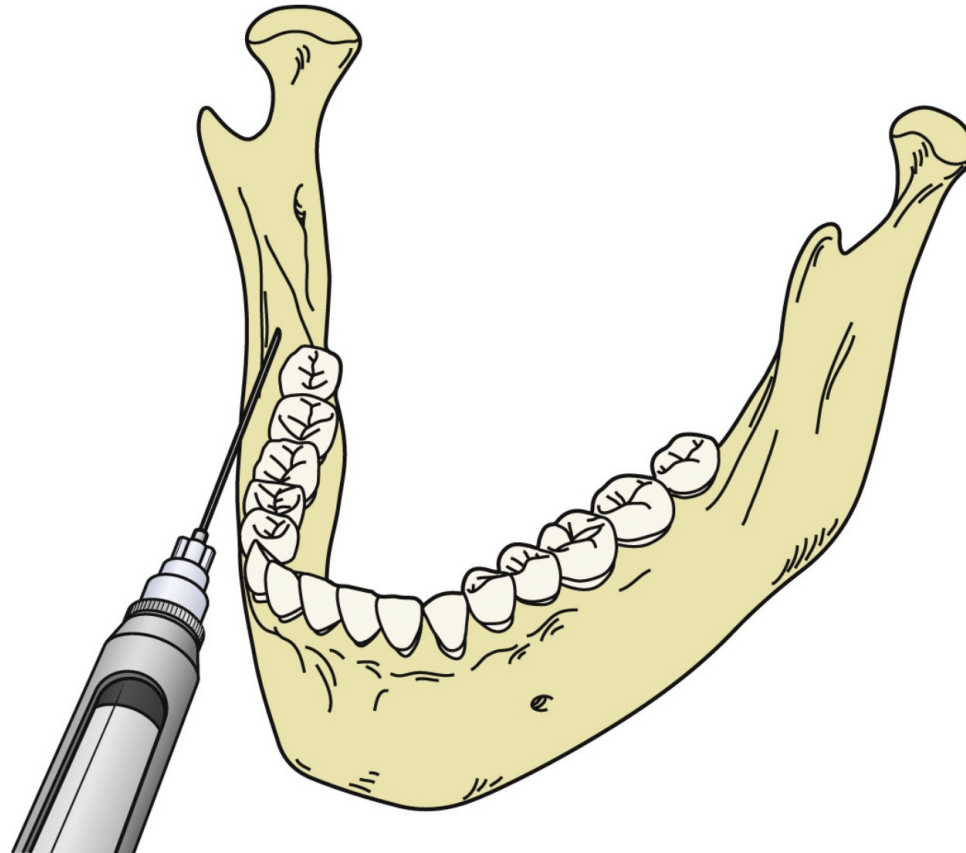








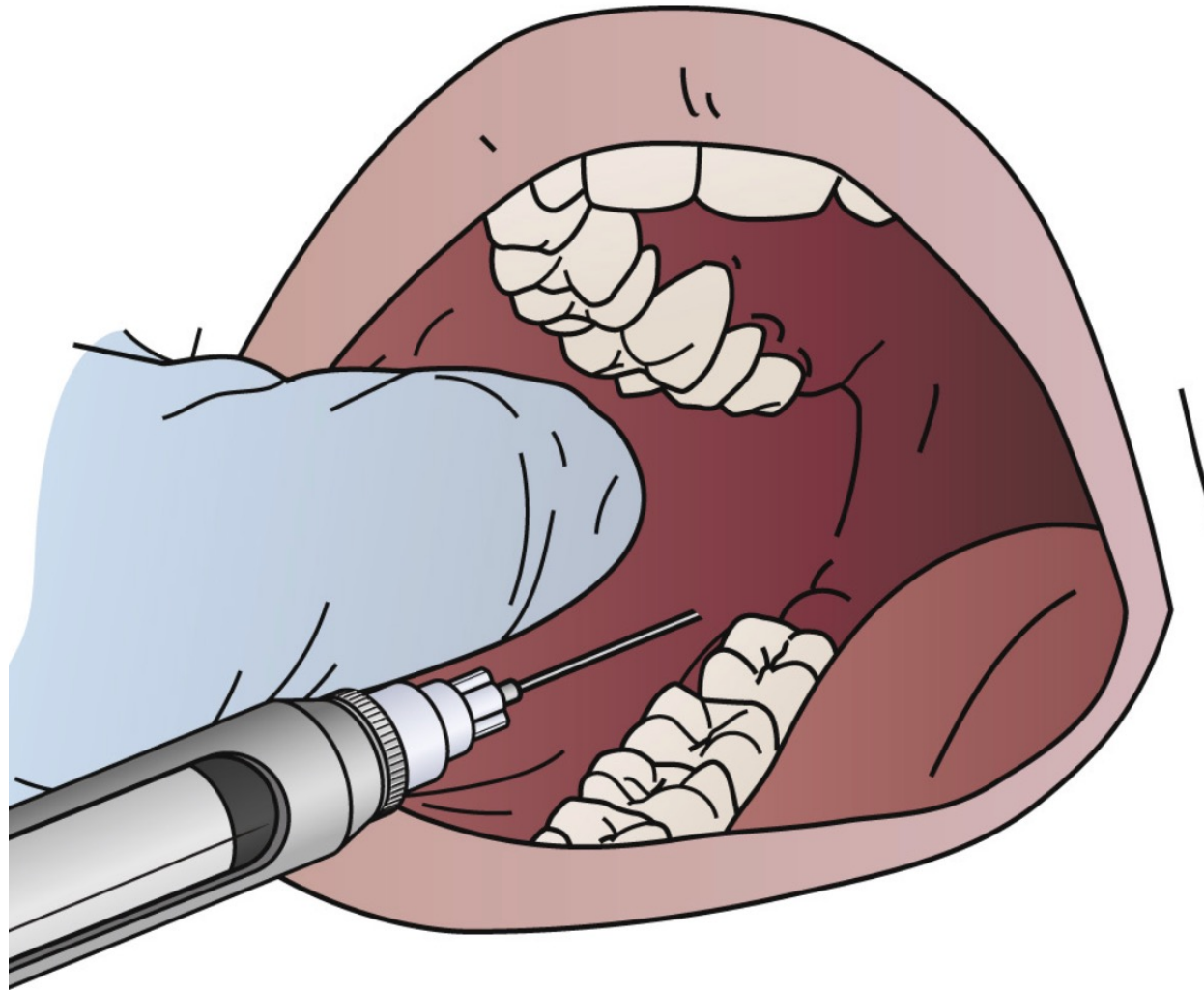
An important point to remember is that in areas of nerve transition, cross-innervation exists. For example, in the region of the mandibular second premolar, the buccal soft tissues are innervated primarily by the mental branch of the inferior alveolar nerve but also by the terminal branches of the long buccal nerve, therefore it is appropriate to supplement an inferior alveolar nerve block with a long buccal nerve block to achieve adequate anesthesia of buccal soft tissue when extracting a lower second premolar or placing an incision into that area.



Periodontal Ligament Injection

Even with profound soft tissue anesthesia and apparent pulpal anesthesia, a patient may continue to have sharp pain as a tooth is luxated. This is especially likely when teeth have pulpitis or if the surrounding soft and hard tissues are inflamed or infected. A technique that should be used in these situations is the periodontal ligament injection. When this injection is delivered properly, with the local anesthetic solution injected under pressure, immediate profound local anesthesia occurs in almost all situations. The anesthesia is short-lived, so the surgical procedure should be one that can be accomplished within 15 to 20 minutes.

Finally, when performing an oral soft tissue biopsy, even when a block technique can provide adequate local anesthesia, infiltrating the tissues adjacent to the planned biopsy site with a vasoconstrictor containing local anesthetic can be useful to help limit bleeding. About 7 minutes should be allowed to elapse between the injection and incision to give the vasoconstrictor time to reach its optimal hemostatic effectiveness.



Managing Postextraction Pain

Although it is self-evident that local anesthesia is necessary for intraoperative pain control, the surgeon should also acknowledge its role in postoperative pain control. For routine extractions for which only mild to moderate analgesics will be needed, usually no additional local anesthetic is necessary. After procedures that have been more traumatic (e.g., the removal of impacted teeth) and where stronger analgesics are likely to be necessary, many surgeons use a long-lasting local anesthetic (e.g., bupivacaine) instead of or in addition to their usual local anesthetic. By doing this, the clinician provides the patient with 4 to 8 hours of local anesthesia.

Anxiety Control

Management of patient anxiety must be a major consideration in oral surgical procedures. Anxiety is a more important factor in oral surgical procedures than in most other areas of dentistry. Patients are frequently already in pain and may be agitated and fatigued, both of which lower the patient's ability to endure pain or pain-producing situations.

Nonpharmacologic Anxiety Control

Anxiety control begins, in most cases, with a proper explanation of the planned procedure, including assurance that the doctor will take all measures to minimize the patient experiencing unexpected sharp pain; an expression of empathy from the dentist of what the patient may be fearing is also appropriate. For the mildly anxious patient with a caring dentist, no pharmacologic assistance, other than LA, is typically necessary for routine extractions.

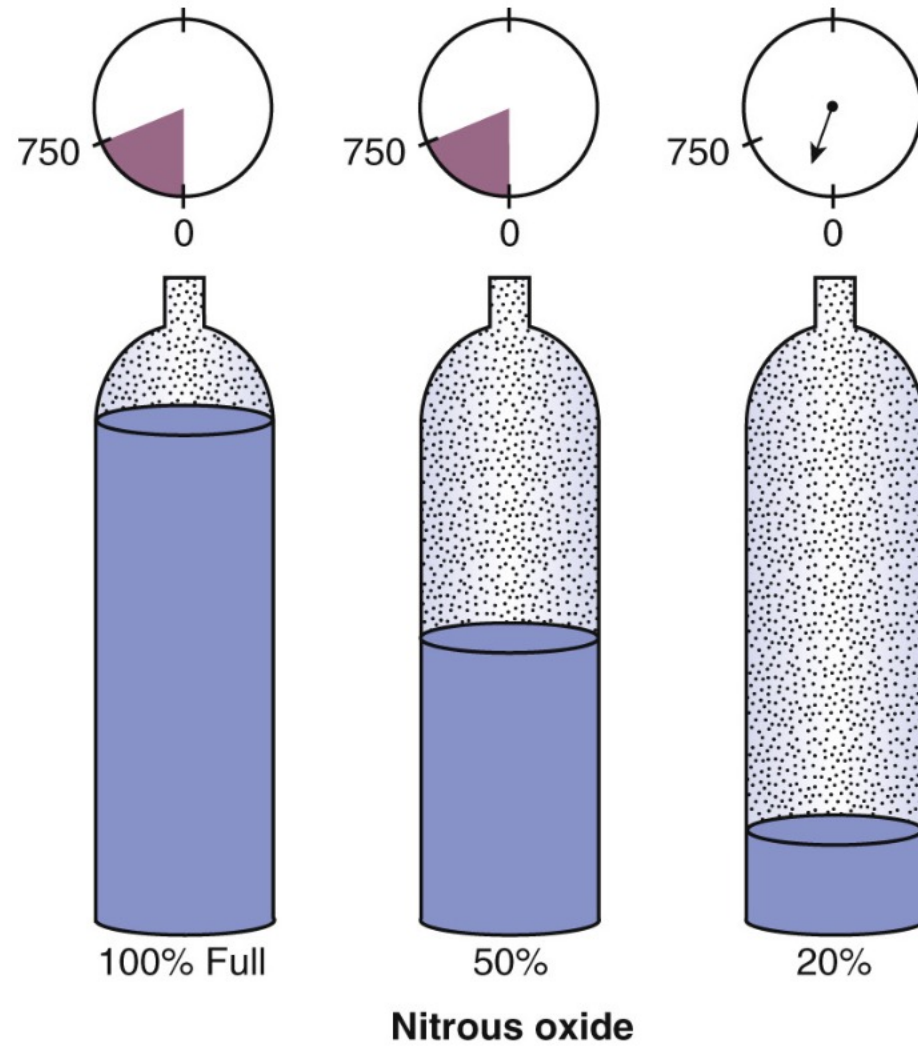
Finally, anxiety is normal even in patients with positive past experiences with extractions because the procedure truly has unpleasant aspects. As noted previously, although the sharp pain is eliminated by local anesthesia, a considerable amount of proprioceptive (pressure) sensation still exists. Other noxious stimuli such as the cracking of teeth and the clinking of instruments are present during an extraction procedure. For these reasons, prudent dentists use a planned method of anxiety control to prepare their patients to deal with the anxiety associated with tooth extraction.

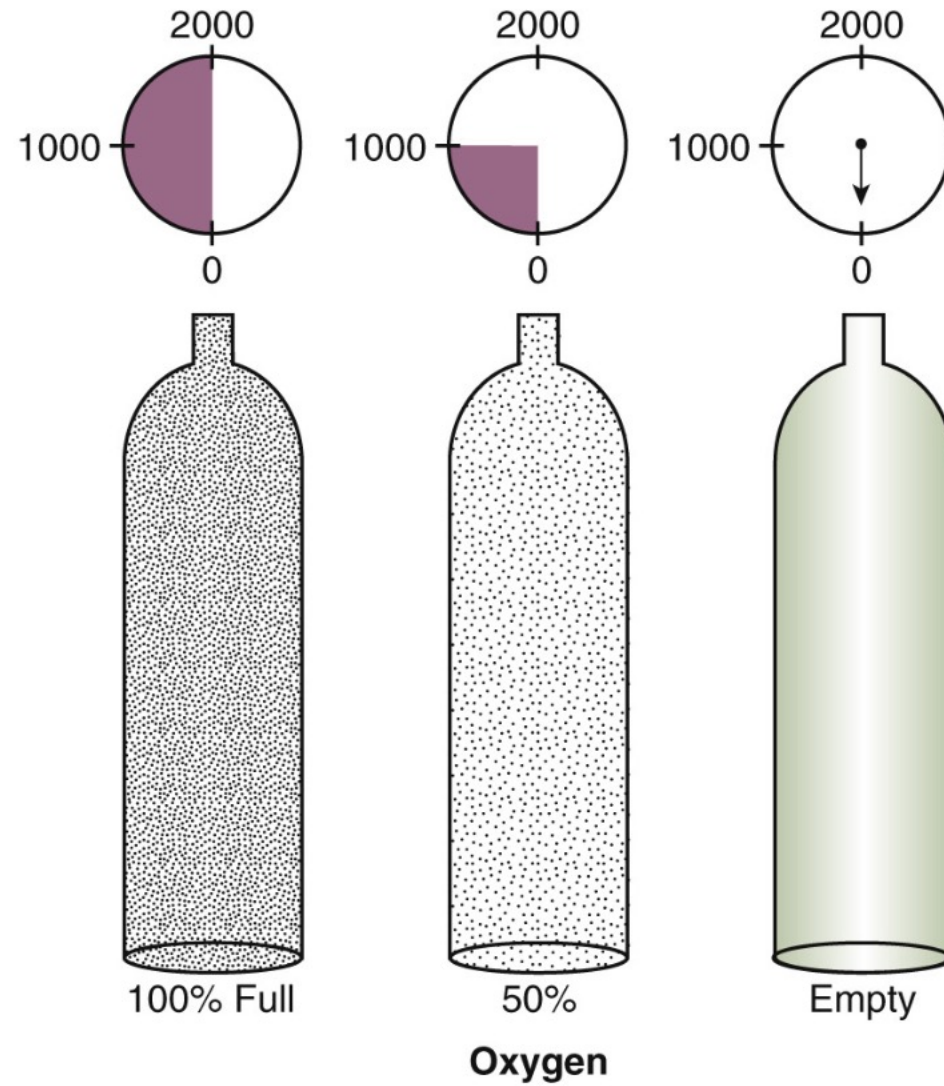
Oral Sedation

As patient anxiety increases, it often becomes necessary to use pharmacologic assistance. Preoperatively, oral medication such as diazepam may help a patient rest well the night before the surgery and provide some relief of anxiety in the morning. A drug such as lorazepam may be useful when administered on the morning of surgery (someone other than the patient should drive him or her to the appointment in such circumstances).

Nitrous Oxide Sedation

Nitrous oxide is an odorless and colorless gas that is not irritating to airways. It is toxic to humans if given in very high concentrations, but, when properly mixed with pure oxygen and given appropriately, it can be an extremely potent analgesic and anxiolytic. This makes it useful when providing oral surgery as well as for patients who primarily fear local anesthetic injections.

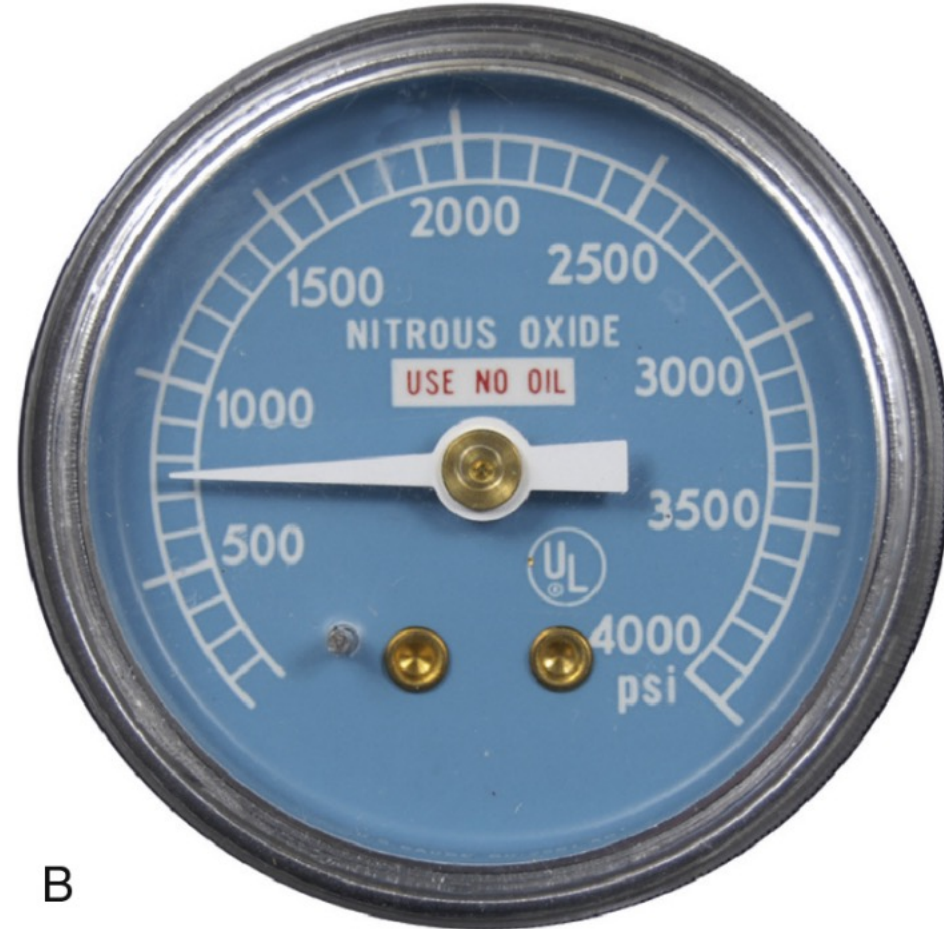




In clinical facilities equipped to deliver nitrous oxide, both the nitrous oxide and oxygen come in pressurized tanks. The primary difference is that nitrous oxide is in the liquid form when under pressure, while oxygen is in the gaseous state under cylinder tank pressure. This results in the differences in readings and behavior of pressure gauges on nitrous oxide and oxygen tanks; while oxygen tank gauges will directly show the actual pressure in the tank at all pressures, nitrous oxide tank gauges will read 750 psi as long as the pressure is above that amount. However, it is only once the pressure falls below 750 psi that the gauge represents the actual tank pressure proportional to the amount of gas remaining



A

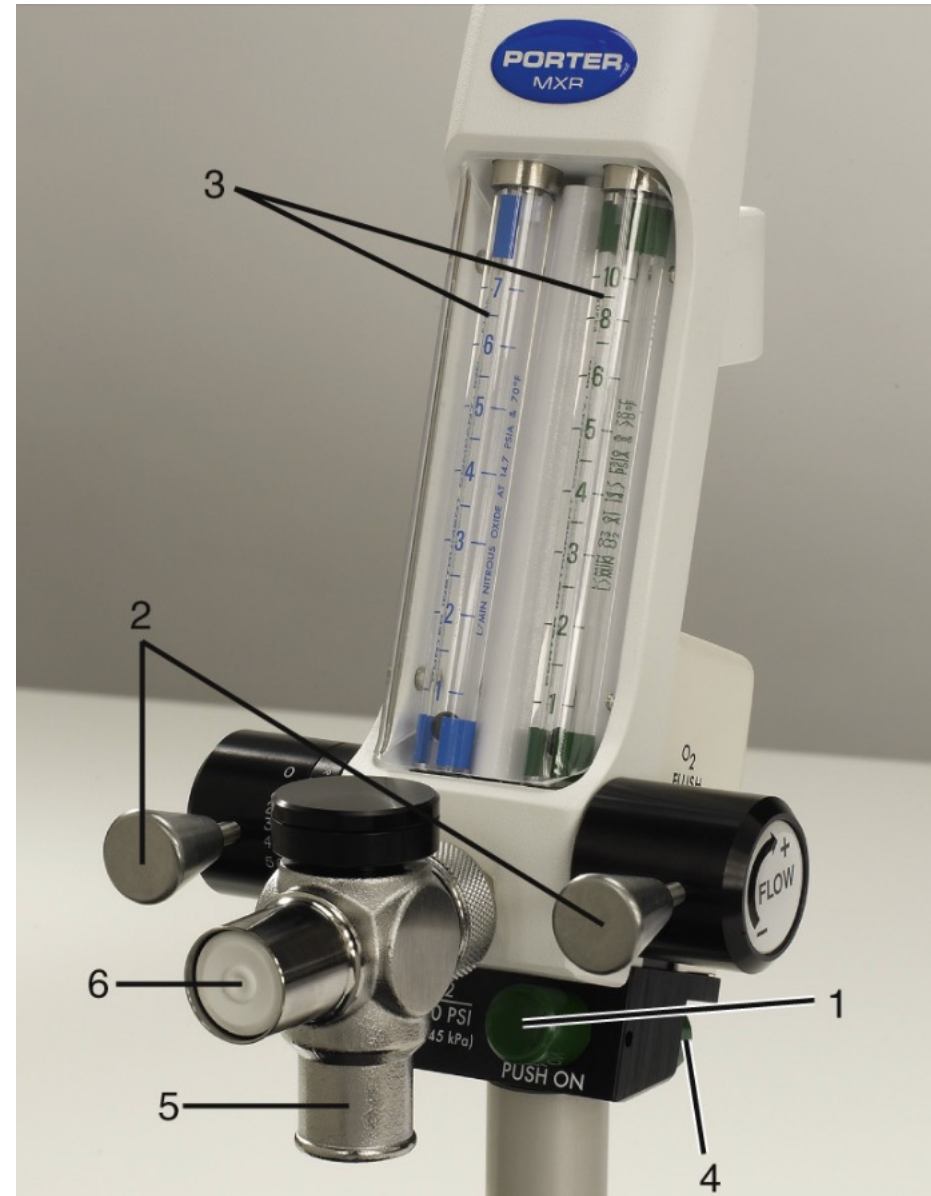


B

Pin index system is used to prevent attaching the wrong gas to the wrong port of the continuous-flow sedation unit/controller. Note the differing patterns of depression below the gas exit port for (A) oxygen and (B) nitrous oxide.









A



B



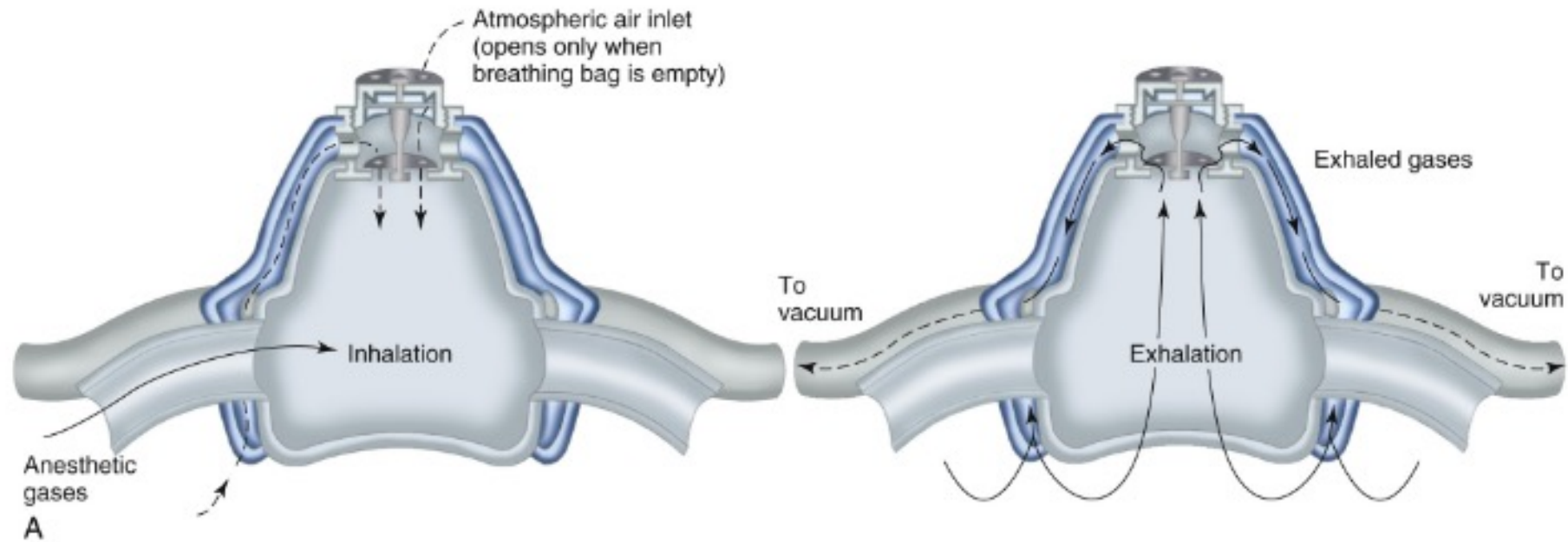
D



C



E





Presedation Evaluation

Before deciding if nitrous oxide should be used with a patient, the doctor must learn about the patient's past medical and anesthetic history. Claustrophobic patients may not be able to tolerate a nasal hood. Patients who cannot handle mind-altering drugs are poor candidates for nitrous oxide; this includes patients with disorders who cannot tolerate the relative loss of control they may feel under the influence of nitrous oxide. Medical problems such as poorly controlled chronic obstructive pulmonary disease or a respiratory infection interfering with nasal breathing are relative contraindications to the use of nitrous oxide sedation. It is prudent to avoid the use of nitrous oxide during the first trimester in pregnant patients. The use in later trimesters should be guided by the patient's obstetrician.



General Protocol for Using Nitrous Oxide

The general protocol for administering nitrous oxide in the dental setting begins with giving the patient 100% oxygen. While this is being done, the clinician should check for proper mask adaptation to the patient's face as well as allow the patient a few minutes to become accustomed to the mask and thus able to tolerate the mask. After a few minutes of preoxygenation, the amount of nitrous oxide (N₂O)/oxygen (O₂) flows should be adjusted to deliver a N₂O:O₂ mix.

Important Caveats to Use of Nitrous Oxide

There are a few important caveats to the use of nitrous oxide sedation. First, like other procedures in dental care, patients must give informed consent to nitrous oxide sedation. Second, like all drugs administered to patients, the dental records must document the patient's vital signs and dose of drug used while undergoing nitrous oxide sedation. This is also useful since once the optimal mixture of nitrous oxide and oxygen is determined. Third, there are rare instances in which patients under the influence of nitrous oxide experience erotic delusions. Therefore a member of the gender opposite that of the operating surgeon should always be present during the use of nitrous oxide. Fourth, someone on the clinical staff should always be present with the patient while nitrous oxide is in use and during the recovery period.



Thank you